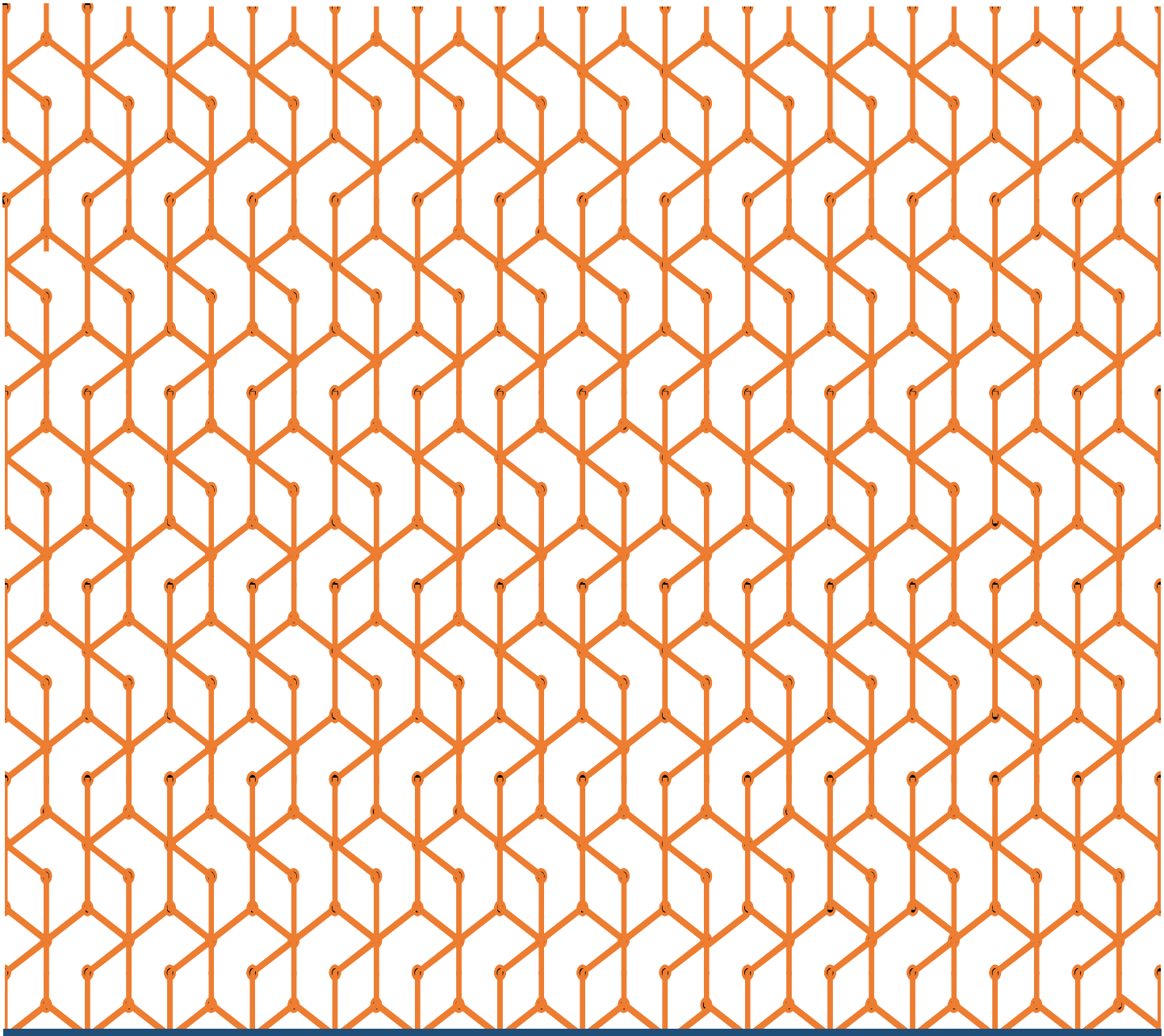

Usability Test for Human Services on Maricopa.gov Website

TWC 421

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4/3/2023

Principles of Web Authoring

Introduction

Maricopa County is the most populous county in the state of Arizona. It's home to the state capital, Phoenix, and almost 4.5 million residents. The official website of the county is Maricopa.gov. The website is a repository of data and links to county services and news. Residents can pay their property tax bills, view air quality ratings, request building permits, apply for a marriage license, etc.

A usability study was created and implemented to determine how easily users can access information and resources for human services/social assistance programs available to them through the county. Participants were provided with a scenario and asked to complete tasks relating to county human services. Tasks were designed to assess the organization of the site and determine whether participants were able to complete each task with ease or difficulty. At the conclusion of the usability test, the researcher determined that the following recommendations will give users a greater chance of successfully completing similar tasks in the future. The recommendations are: prioritize use of the search bar, group links to all human services together throughout the site, and add a GET HELP WITH category with comprehensive subheadings in the "I Want To…" hover navigation.

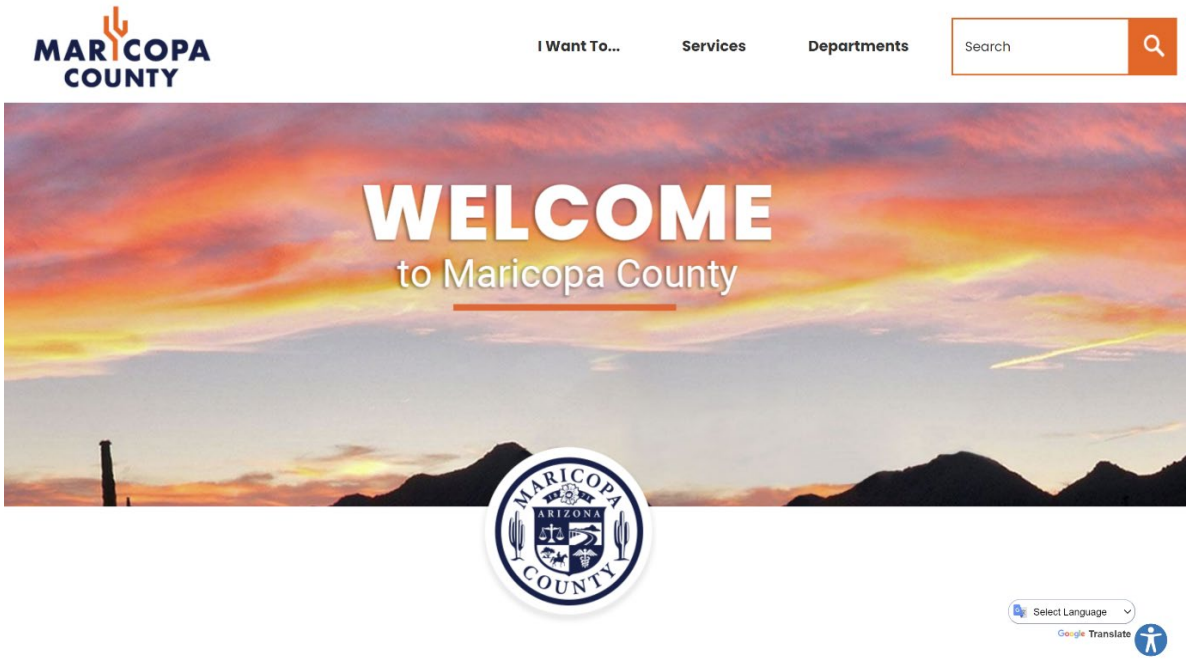

Figure 1: Maricopa.gov homepage.

Methodology

This study was conducted using three participants with varying ages and technical skills. The only criteria for participants were that they have experience using a mouse and browsing webpages on a PC or laptop. The participants consisted of a 16-year-old male, 73-year-old female, and 41-year-old female.

Table 1: Participant Demographics.

Participant	Age	Gender	Education level	Employment
#1	16	Male	High school	None
#2	73	Female	Some college	Retired
#3	41	Female	B.A. degree	Full-Time

Participants completed the study in person at a table in the researcher's home. Each participant was tested separately using the researcher's laptop (Dell Inspiron 16 Plus running Windows 11 Home) and wireless mouse (Logitech M510). The researcher sat to the left of each participant with printouts of a Welcome Script (Appendix A), a blank Entry Questionnaire (Appendix B), a Background Scenario/tasks printout (Appendix F), and a blank Exit Questionnaire (Appendix G). The researcher also used their phone to record audio of each test, with the participant's permission. The laptop displayed the Google Chrome search page and nothing else.

Once seated, the researcher read through the Welcome Script (Appendix A). The researcher then asked the participant about their website usage and favorite websites to make the participant comfortable answering questions. The researcher then started their phone recorder and asked the participant to type Maricopa.gov into the Google Omni bar. Once the website was visible the researcher asked the participant to scroll through the website's homepage and look around without clicking on anything. After the participant spent 2-3 minutes viewing the homepage, the researcher verbally administered the Entry Questionnaire (Appendix B) and recorded the participant's answers (Appendices C-E).

At the conclusion of the Entry Questionnaire, the researcher introduced the Background Scenario and four tasks (Appendix F). The scenario and task #1 were read aloud to the

participant, they would attempt to complete it, and the researcher would read aloud the next task, repeating the pattern for all four tasks. When a participant completed a task, the researcher would move on to the next, but when a participant struggled too much, the researcher also moved on to the next task. The decision to move to the next task before the current one was complete was based on the frustration level of the participant. Behaviors that prompted the researcher to move on were vocalization of frustration through sounds or words, rapid, angry mouse clicking, and cessation of interaction with the site. The researcher followed Krug's (2010) recommendations regarding miserable and/or frustrated participants (p. 117).

Upon completion or attempted completion of all four tasks, the researcher verbally administered the Exit Questionnaire (Appendix G) to each participant and recorded their answers (Appendices H-I).

Results

Observations, review of test recordings, notes, and answers to Exit Questionnaires (Appendices H-J) lead the researcher to believe that completion of the tasks was somewhat difficult for all three participants as none managed to complete more than two tasks. Below is a table summarizing task results, followed by a more in-depth breakdown:

Table 2: Participant Task Results.

	Participant		
Tasks	#1	#2	#3
#1	✓	✗	✓
#2	✗	✓	✗
#3	✗	✗	✗
#4	✓	✓	✓

Task #1

You are having a difficult time affording groceries for the month. Is there information on the Maricopa.gov website that will help you make sure you have enough food?

Participant #1

Completed task in under a minute. Hovered on I Want To…, then Most Visited Departments, Human Services>Help with Bills>Connect to Food Assistance.

Participant #2

Was unable to complete task and researcher moved on when participant vocalized for the second time, "This shouldn't be this hard." Participant hovered on I Want To…, scrolled down the homepage to Most Visited Departments, Human Services>Senior/Adult Services, and found a 404 error.

Participant #3

Took over three minutes to complete the task. Path: Hovered on I Want To…, then Departments>Human Services>Community Services>Departments>Public Health>Homepage>Human Services>Senior and Adult Services>Homepage>Human Services>Apply for Help>Connect to Food Assistance. Participant often sighed and made the following comments: "They aren't making this easy" and "I'm just going to guess here."

Task #2

You would like some help dealing with stress and anxiety. Is there information on the site to assist you with mental health issues?

Participant #1

Was unable to complete the task. Attempted similar path for task #1, then went back and selected Public Health rather than Human Services. Stated, "I thought it would be here."

Participant also tried the “I Want To…” hover box then Human Services again and declared, “I’m at a loss.”

Participant #2

Task was completed in under 30 seconds. Path: Services>Free/Low-Cost Health Services>Mental Health/Substance Use Disorders.

Participant #3

Not completed. Used same path as task #1 when they found food assistance. When they did not see mental health in the same place, they stated they were done looking.

Task #3

Your boss told you that if you improve your computer skills, they will increase your pay. Locate any info on the site that would help you with job training.

Participant #1

Completed task but did not realize it due to confusion over “Apply” button and lack of hyperlinks as shown in Figure 2 below.

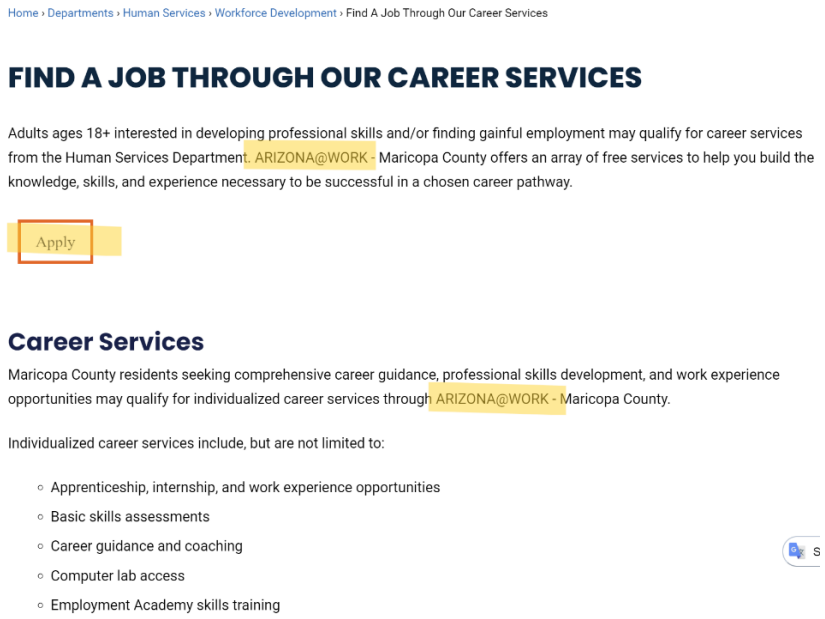

Figure 2: Maricopa.gov Career Services page.

Participant #2

Did not complete the task due to frustration but did find the same 404 error from task #1. Path: Most Visited Departments, Human Services>Senior/Adult Services. Participant stated that they didn't want to look anymore if they couldn't use the search bar.

Participant #3

Completed the task in under 30 seconds. Path: Departments>Human Services>Workforce Services. Participant wondered if there was online training for people without transportation but was not able to answer their own question based on what they saw on the Workforce Services page.

Task #4

Due to an unexpected expense, you aren't sure you'll be able to pay your rent next month. Is there a way to apply for rent assistance online?

Participant #1

Completed task in under 30 seconds. Path: Most Visited Departments, Human Services>Help with Rent & Utilities.

Participant #2

Completed task in under 30 seconds. Path: Services, Help Paying Bills>Rent and Mortgage Assistance.

Participant #3

Completed task in under 30 seconds. Path: Departments, Huma Services>Help with Rent & Utilities.

Recommendations

The researcher has three recommendations to improve the usability of the Maricopa.gov website. The recommendations are based on the researcher's observations of participants' verbal and non-verbal output during the test cycle. Recommendations were also chosen for their simplicity. "What's the smallest, simplest change we can make that's likely to keep people from having the problem we observed?" (Krug, 2010, p. 167).

Prioritize the Use of the Search Bar

All three participants said that they would use the search bar rather than comb through a website. Participants #1 and #2 said this twice. The researcher recommends moving the search bar from the upper right-hand corner to just below the "Welcome" text on the home page. Beneath the search box, include an explanation such as: Type the service or term/s you are looking for into the search bar above. None of the participants were able to complete task #3: locate any info on job training. As a quick experiment, the researcher typed "job training" into the search bar and received an excellent result in 0.16 seconds (see Figure 3).

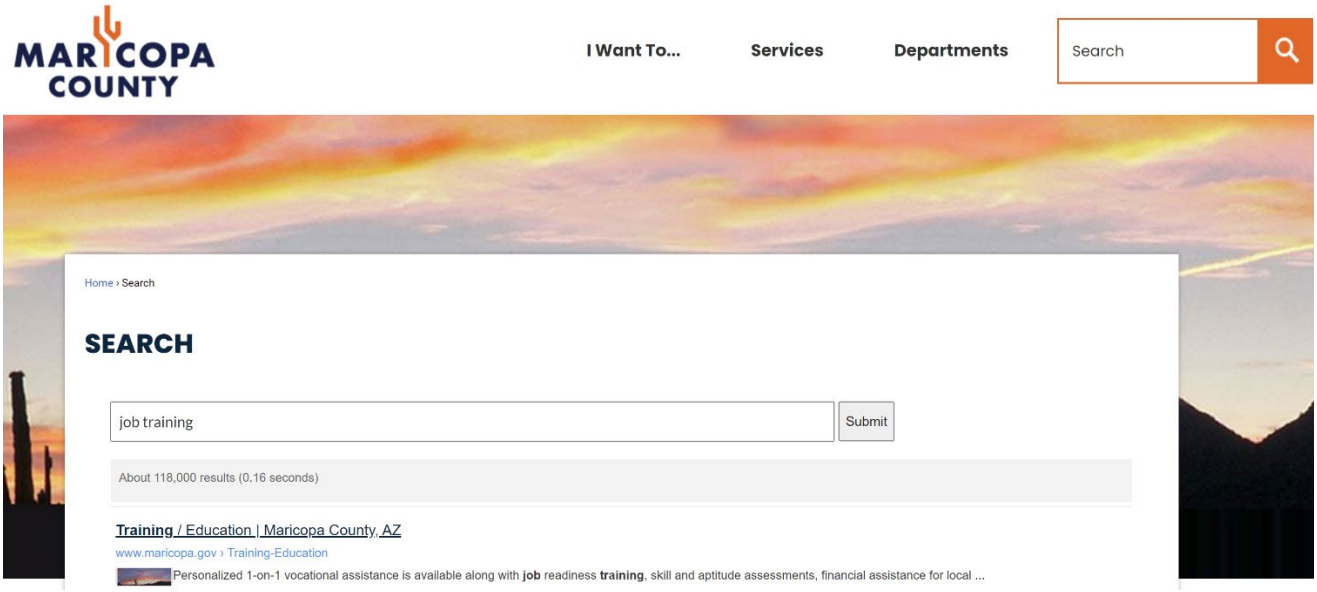

Figure 3: Maricopa County job training search bar results.

Group Links to Human Services Together

The researcher noted that participants would begin a task and say that they thought they had already found that service while working on a previous task. They either misremembered or couldn't remember how they found it the first time as many similar services were included on some pages but not others. There is a page that contains a link to all Human Services, but it is very difficult to find. Path: Departments>Human Services>Apply for Help>Other Assistance Programs>All Services and Support>Maricopa County Rescue Service Programs. This page, https://www.maricopa.gov/5774/Maricopa-County-Rescue-Service-Programs, should be the page users see whenever Human Services is clicked upon.

Add Subcategory to I Want To… Navigation Category

There are only three navigation categories on the home page: I Want To…, Services, and Departments. The researcher noted that during the first test, all three participants were immediately drawn to the I Want To… navigation, which doesn't contain any human services headings/links. To address this, the researcher proposes to add a heading/link entitled GET HELP WITH... Under this new heading would be subheadings/links for all the human services listed on the Maricopa County Rescue Service Programs page.

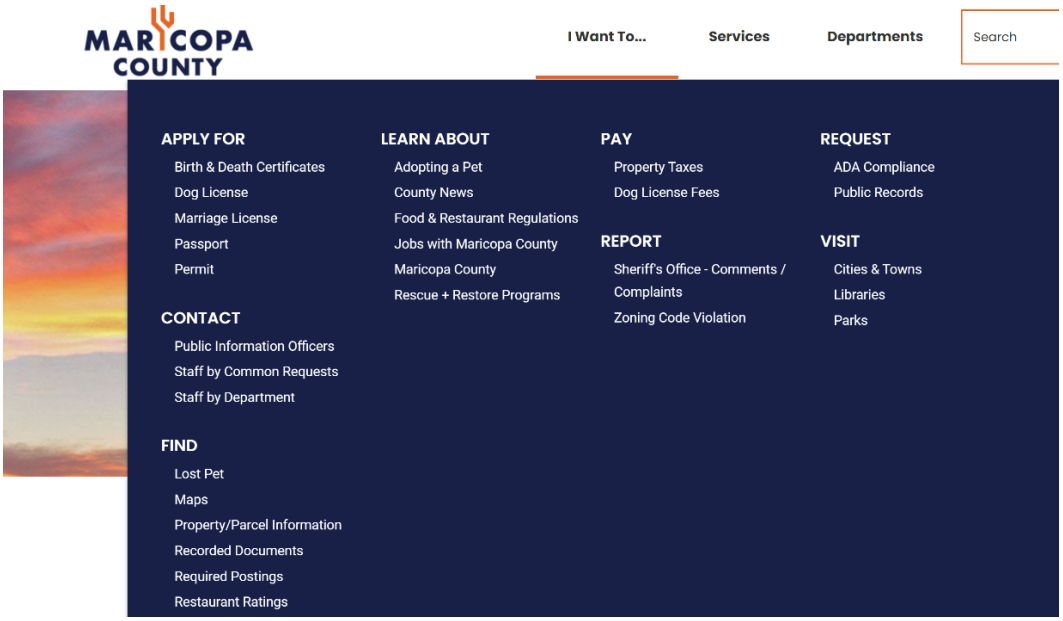

Figure 4: I Want To... hover navigation.

References

Maricopa County, AZ | Official Website. (n.d.-b). https://www.maricopa.gov/

Krug, S. (2010). *Rocket Surgery Made Easy: The Do-it-yourself Guide to Finding and Fixing Usability Problems*. New Riders Pub.

Appendix A

Welcome Script

Hi, I'm going to be walking you through this session today.

I'm asking people to try using a website that I'm working on so I can see whether it works as intended. The session should take around 30-45 minutes.

The first thing I want to make clear is that I'm testing the *site*, not you. You can't do anything wrong here and there are no wrong answers.

As you use the site, I'm going to ask you to try to think out loud: to say what you're looking at, what you're trying to do, and what you're thinking.

Please don't worry that you're going to hurt my feelings. I'm doing this to improve the site, so I need to hear your honest reactions.

If you have any questions just ask them, but I may not be able to answer right away, since we're interested in how people do when they don't have someone helping them. If you need to take a break at any point, just let me know.

With your permission, I'm going to record what happens on the screen and our conversation. The recording will only be used to help me figure out how to improve the site, and it won't be seen by anyone else.

Before we begin the test, I will have a short-entry questionnaire for you to complete. I will then observe you complete four tasks on the site, then we'll wrap up with an exit questionnaire. Do you have any questions so far?

Appendix B

Blank Entry Questionnaire

Please look at the homepage of Maricopa.gov. Scrolling is fine, but please don't click on anything.

1. What is the purpose of the site?
2. What would you use the site for?
3. What do you think about the design and organization of the site?

Appendix C

Completed Entry Questionnaire #1

In the address bar please type in Maricopa.gov. Go ahead and take a few minutes to scroll through the page and look around, but please don't click on anything.

1. What is the purpose of the site?

   It has news and things to do in Maricopa County.

2. What would you use the site for?

   To keep updated about Maricopa County.

3. What do you think about the design and organization of the site?

   Pretty good. Easy to see, nothing too small, very readable.

Appendix D

Completed Entry Questionnaire #2

In the address bar please type in Maricopa.gov. Go ahead and take a few minutes to scroll through the page and look around, but please don't click on anything.

1. What is the purpose of the site?

   To do almost anything you need to do in Maricopa County: pay taxes, permits, restaurant ratings, maybe jury duty.

2. What would you use the site for?

   Pay taxes, driver's license renewal, or plate or vehicle registration. I'd look at the restaurant ratings.

3. What do you think about the design and organization of the site?

   I like it better than the old one. Looks new. I can see more services on one site. I like the big font.

Appendix E

Completed Entry Questionnaire #3

In the address bar please type in Maricopa.gov. Go ahead and take a few minutes to scroll through the page and look around, but please don't click on anything.

1. What is the purpose of the site?

   Access county services and news.

2. What would you use the site for?

   To look up county codes and documents.

3. What do you think about the design and organization of the site?

   Why does the name of the site take up the whole screen? There are so many menus and lists. It's too much.

Appendix F

Background Scenario

You are experiencing some increased stress and worry about school and/or work, and you'd like some help dealing with it, but do not have health insurance. Your job also does not pay very much. As you live in Maricopa County, Arizona, specifically the city of Chandler, you wondered if the county would have any programs that could assist you.

Instructions: Please complete the following tasks without using the search bar. Remember to take your time and to think out loud while you are completing them.

Task #1

You are having a difficult time affording groceries for the month. Is there information on the Maricopa.gov website that will help you make sure you have enough food?

Task #2

You would like some help dealing with stress and anxiety. Is there information on the site to assist you with mental health issues?

Task #3

Your boss told you that if you improve your computer skills, they will increase your pay. Locate any info on the site that would help you with job training.

Task #4

Due to an unexpected expense, you aren't sure you'll be able to pay your rent next month. Is there a way to apply for rent assistance online?

Appendix G

Blank Exit Questionnaire

1. What did you like about this website?

2. What did you not like?

3. How hard or easy was it for you to complete the tasks?

4. What would you change?

Appendix H

Completed Exit Questionnaire #1

1. What did you like about this website?

   It's easy to read. There's a lot of information on the homepage. I have a good idea of what stuff is offered.

2. What did you not like?

   I'm not used to looking for stuff because I always use the search bar. I expected to find stuff and didn't.

3. How hard or easy was it for you to complete the tasks?

   The food task was easy. I never found mental health stuff. I couldn't find job training. The rent assistance one was easy.

4. What would you change?

   Nothing. I would just use the search bar instead.

Appendix I

Completed Exit Questionnaire #2

1. What did you like about this website?

   It's easy to use. It's easy to find the things I use. I like the large font. I like the "Common Requests" section.

2. What did you not like?

   The white background is too bright; it hurts my eyes.

3. How hard or easy was it for you to complete the tasks?

   I got two out of four. Making food hard to find is a major flaw.

4. What would you change?

   Food is a major thing. The first things people need are food and medical. There was more stuff for animals than people on the home page.

Appendix J

Completed Exit Questionnaire #3

1. What did you like about this website?

   The top navigation bar and the colors.

2. What did you not like?

   The use of all caps for the departments and the organization.

3. How hard or easy was it for you to complete the tasks?

   Somewhat difficult. There was a lot of guessing.

4. What would you change?

   Organize Human Resources better. Make it make sense, so you don't have to look all over for the same types of services.